Step 1: Geometry Generation

The disc is defined by **11 parameters** in mm, all stored in NOMINAL_GEOMETRY_MM in config.py :

Parameter	**Nominal value**
bore_radius_inner	24.0 mm
bore_height	11.0 mm
bore_thickness	30.0 mm
lower_transition_height	8.0 mm
web_height	40.0 mm
web_thickness	10.0 mm
upper_transition_height	9.0 mm
rim_height	15.0 mm
rim_thickness	20.0 mm
lower_fillet_radius	2.2 mm
upper_fillet_radius	2.6 mm

One hard constraint is enforced: bore_thickness > rim_thickness > web_thickness (30 > 20 > 10) . This is the classic I-beam-like turbine disc cross section.

From those parameters, **6 radial stations** are computed by stacking the heights outward: $r_0 = 24, r_1 = r_0 + 11, r_2 = r_1 + 8, r_3 = r_2 + 40, r_4 = r_3 + 9, r_5 = r_4 + 15$. These define the zone boundaries.

build_disc_contour() in geometry.py then constructs the **closed 2D boundary polygon** of the meridional cross-section . It does this by:

1. Sampling 220 points along the front surface (axial coordinate = $-t(r)/2$) and 220 along the rear ($+t(r)/2$), where $t(r)$ is the thickness profile
2. Closing the polygon with 20-point caps at the inner bore face and outer rim face
3. Validating the contour is not self-intersecting

The thickness transition between zones uses a **fillet blend function** — a smooth power-law interpolation whose sharpness depends on the fillet radius relative to the thickness step . Smaller fillet radius → sharper shoulder.

Step 2: "FEA" — What's Actually Happening

Important caveat: this is not a general-purpose FEA solver. There is no assembled stiffness matrix or incremental time integration. It's an **analytical stress surrogate** encoded in physics.py that mimics what FEA would produce for a rotating disc, computed directly at mesh nodes using closed-form expressions .

There are also **no time steps** in the traditional sense. Instead, the simulation uses **7 flight-cycle phases** as discrete load cases :

Phase	Speed factor	Weight
taxi	0.20	0.20
takeoff	1.00	0.08
climb	0.86	0.15
cruise	0.78	0.32

Phase	Speed factor	Weight
descent	0.55	0.12
reverse_thrust	0.46	0.05
taxi_return	0.18	0.08

The speed factor scales the centrifugal load (stress ∝ ω^2), so stress at each phase = base stress × speed_factor^2 .

How stress is computed

compute_phase_equivalent_stresses() builds the stress field node-by-node as a product of several terms :

1. **Rotor shape** — a blend of radial and hoop stress components, both increasing with normalized radius: 0.44 × radial_term + 0.56 × hoop_term
2. **Thin-section amplification** — nodes in thinner sections get higher stress, using the ratio of a global reference thickness to local thickness raised to power 0.72
3. **Transition stress concentration** (_transition_concentration) — Gaussian peaks at the shoulders of the lower and upper transition zones; inversely proportional to fillet radius (smaller fillet → higher K_t)
4. **Bore concentration** — adds stress at the bore shoulder, scaled by bore slenderness ratio (height/thickness relative to nominal)
5. **Web concentration** — adds stress at mid-web, scaled by web slenderness
6. **Zone multiplier** — transitions (zones 1 and 3) get a flat ×1.04 boost
7. **Region scale** — bore ×1.10, web ×1.00, rim ×1.07
8. **Global geometry gain** — a small linear term based on rim thickness + web height + bore radius

The final result is a (N_nodes × 7) array of stresses — one column per flight phase .

Step 3: Life Calculation

Life is computed in compute_life_raw() using a **Palmgren-Miner linear damage accumulation** across all 7 phases .

Stress amplitude

First, the phase stresses are converted to stress amplitudes. Each phase gets an amplitude scale factor:

$$\sigma_a = \sigma_{eq} \times 0.5 \times (0.85 + 0.15 \times \text{speed\_factor})$$

This encodes a mean-stress-like correction — takeoff (speed=1.0) has a higher amplitude contribution than taxi .

Geometry life severity

Before entering the S-N curve, stresses are further multiplied by a **local geometry severity** factor from _geometry_life_severity() . This accounts for:

- Thickness ratio relative to nominal (thinner = more severe)
- Zone-specific contributions: bore slenderness, fillet radii at transitions, web slenderness, rim thinness
- Clipped to [0.65, 2.85]

S-N curve (zone-specific, piecewise log-log)

Each zone has its own S-N law with a **knee point** :

Zone	Knee stress (MPa)	Knee life (cycles)	Slope above knee	Slope below knee
bore	60	20,000	9.1	4.0
lower_transition	77	65,000	9.2	4.1
web	104	150,000	7.6	3.3
upper_transition	107	210,000	8.4	3.7

Zone	Knee stress (MPa)	Knee life (cycles)	Slope above knee	Slope below knee
rim	108	200,000	7.8	3.4

If $\sigma_a \geq \sigma_{knee}$: $N = N_{knee} \cdot (\sigma_a/\sigma_{knee})^{-m_{high}}$, otherwise the shallower slope m_{low} applies .

Miner's rule

Damage per flight cycle is summed across all 7 phases:

$$D = \sum_p \frac{w_p}{N_{fail,p}}$$

where w_p is the phase weight. Life is then $1/D$, computed per node . The minimum life across all nodes is what you'd typically report as the disc life.